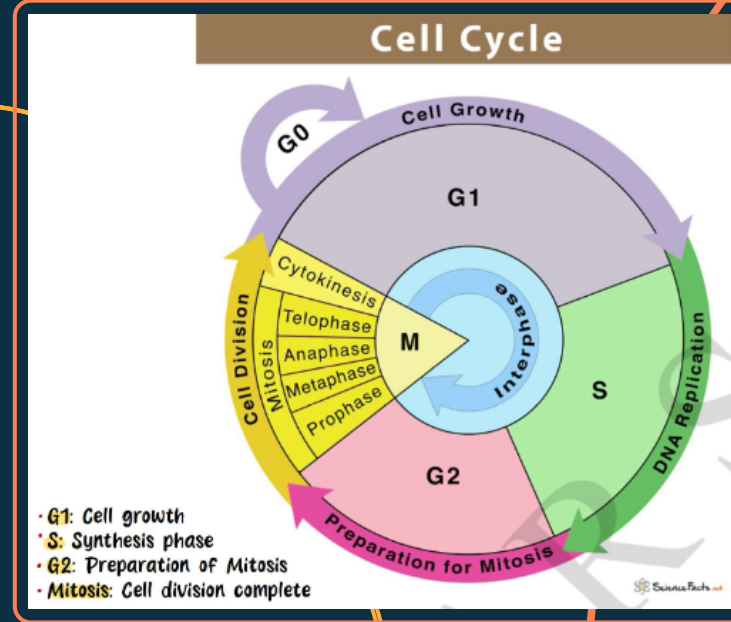




Definition

- Study of Cell
- Building blocks of life
- Fundamental unit of life
- Structural or functional basis of life

Discovery

- 1st by Robert Hooke (1665)
 - in cork ((Dead Cell))
 - Father of cytology
- by Antonie van Leeuwenhoek (1674)
 - in pond water (Living Cell)
- by Robert Brown (1831)
 - Nucleus
- by Fredrick Mischer (1869)
 - Molecule discovery of DNA
- Watson & Crick (1953)
 - Double-helical structure of DNA
- by Camillo Golgi
 - Golgi bodies

Cell Theory

- All the plants & animals are composed of cells
- By Schleiden & Schwann (1838)
- Omnis Cellula E Cellula
 - by Rudolf Virchow
 - New cells are formed from the pre-existing cells

Cell cycle

Types of cell

Prokaryotic Cell	Eukaryotic Cell
- Size: 1-10 μm - Nucleus absent - Nucleolus absent - Cell organelles absent - Chromosome: single	- Size: 5-10 μm - Nucleus present - Nucleolus present - Cell organelles present - Chromosome: more than one
Example: Bacteria, Cyanobacteria (Blue green algae), mycoplasma	Example: Plant and Animal cell

Cell membrane

- Outermost covering of the cell
- Thin, elastic, flexible, Living
- provide definite shape & holds content
- Selectively permeable
- Proteins + lipids (Composition)
- Diffusion
 - Spontaneous movement of molecules from high to lower concentration
- Osmosis
 - Movement from high to low but through a semipermeable membrane
 - Hypertonic - Solute > Solvent
 - Hypotonic - Solute < Solvent
 - Isotonic - Solute = solvent
- Cell wall
 - Provides determines shape, strength & turgidity
 - Freely permeable, Non-living
 - Plants - made up of Cellulose
 - Bacterial - made up of Peptidoglycan/Peptin
 - Fungi - made up of Chitin

Nucleus

- Largest cell structure
- "Brain of the cell"
- Nuclear envelope
 - Composed of membranes & Covers Nucleus
 - Separates nucleus from cytoplasm & permits materials between them
- Nucleoplasm
 - Liquid ground substance of nuclear envelope
 - Nucleolus
 - one or more in number
 - Ribosomes - forms protein forming centres
 - Chromatin material
 - Entangled mass of thread like structures
 - During cell division they get organised in chromosome

Chromosome

- Thread like structures & visible during cell division
- Components
 - DNA
 - Consists of genes
 - Proteins
- 23 Pairs
 - 22 pairs (autosomal)
 - 1 pair (sex pair - XX & XY)
- Cytoplasm
 - Parts between plasma membrane & nuclear envelope
 - Cell organelles present in it
 - Cytosol - Aqueous ground substance
 - Cytoplasm Inclusions - Non-living component in cytoplasm eg. Pigments, nutrients etc

Plastids

- Only in plant cell
- Has its own DNA & Ribosomes
- Types
 - Chloroplasts
 - stores chlorophyll pigment
 - Chromoplasts
 - stores colour pigment
 - Leucoplasts
 - Amyloplast - Storage of Carbohydrates
 - Aleuroplast - Storage of Proteins
 - Elaioplast - Storage of Fats

Cell organelles

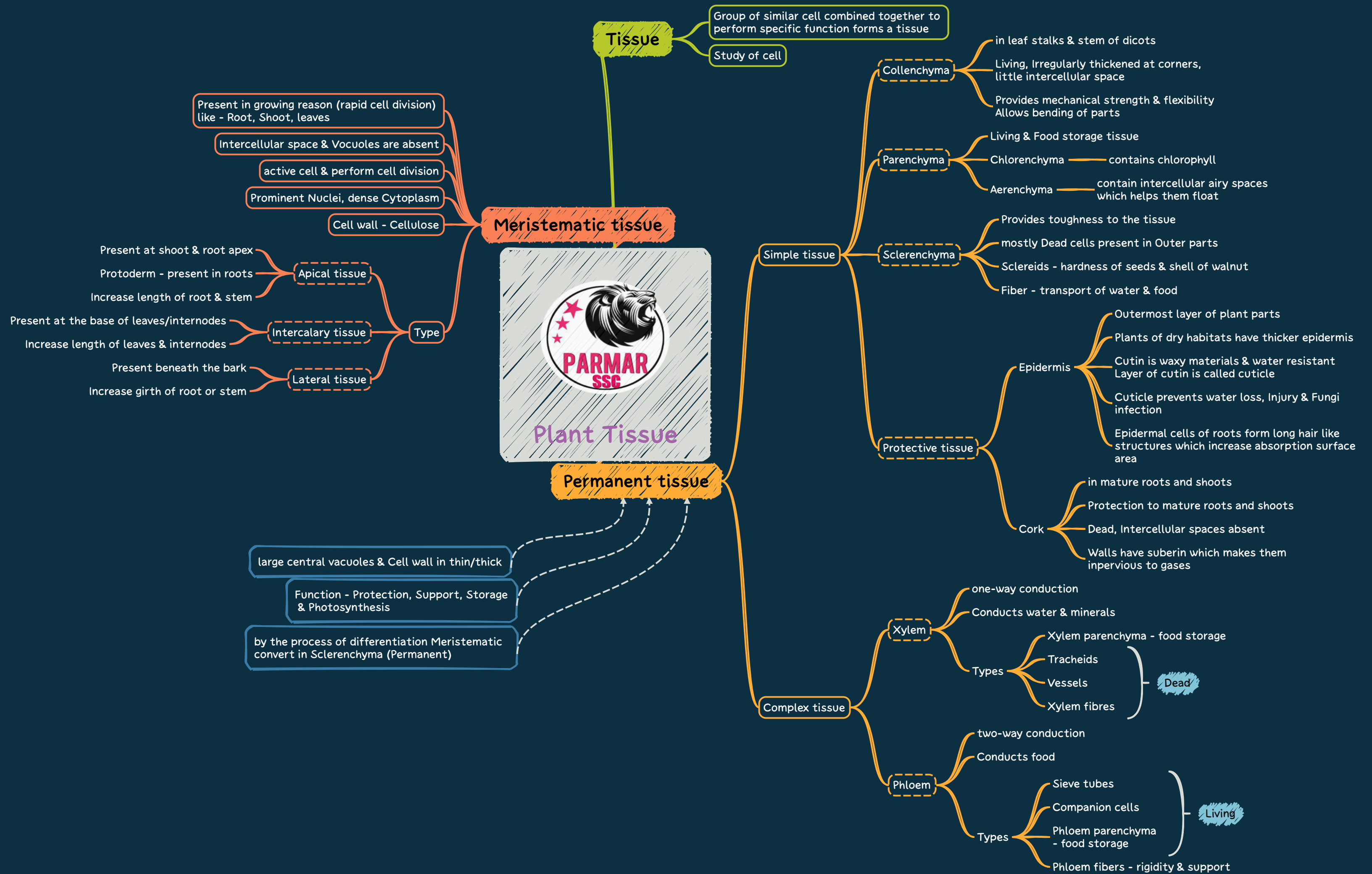
- Mitochondria
 - Powerhouse of the cell
 - Sausage shape
 - Energy currency - ATP (Adenosine Triphosphate)
 - Two-membranes
 - Outer - Porous
 - Inner - ATP synthesis
 - Has its own DNA & Ribosomes
 - involved in Apoptosis
 - Suicidal bag of the cell
- Lysosomes
 - Filled with Digestive enzyme
 - breakdown long chain of Fatty acids through beta oxidation
- Peroxisome
 - Neutralises harmful toxins
- Golgi bodies
 - Cis face - towards nucleus (Convax)
 - Trans face - towards plasma membrane (Concave)
 - Functions
 - Packaging & Dispatching
 - Stores, modification of package
 - Forms lysosomes
 - Types
 - Rough ER
 - Synthesis of proteins, digestive enzyme
 - ribosomes present
 - Smooth ER
 - synthesis of fats & lipids, Detoxify poisons & Drugs
 - ribosomes absent
 - Long Tubules, Vesicles & Cisternae
- Vacuoles
 - in plants
 - Large, distinct, Permanent
 - Provide turgidity & rigidity
 - Store cell sap (consist of - Amino acids, Sugars, Organic acid)
 - in animals
 - 50-90% of cell
 - Small, temporary
 - stores glucose, glycogen, protein

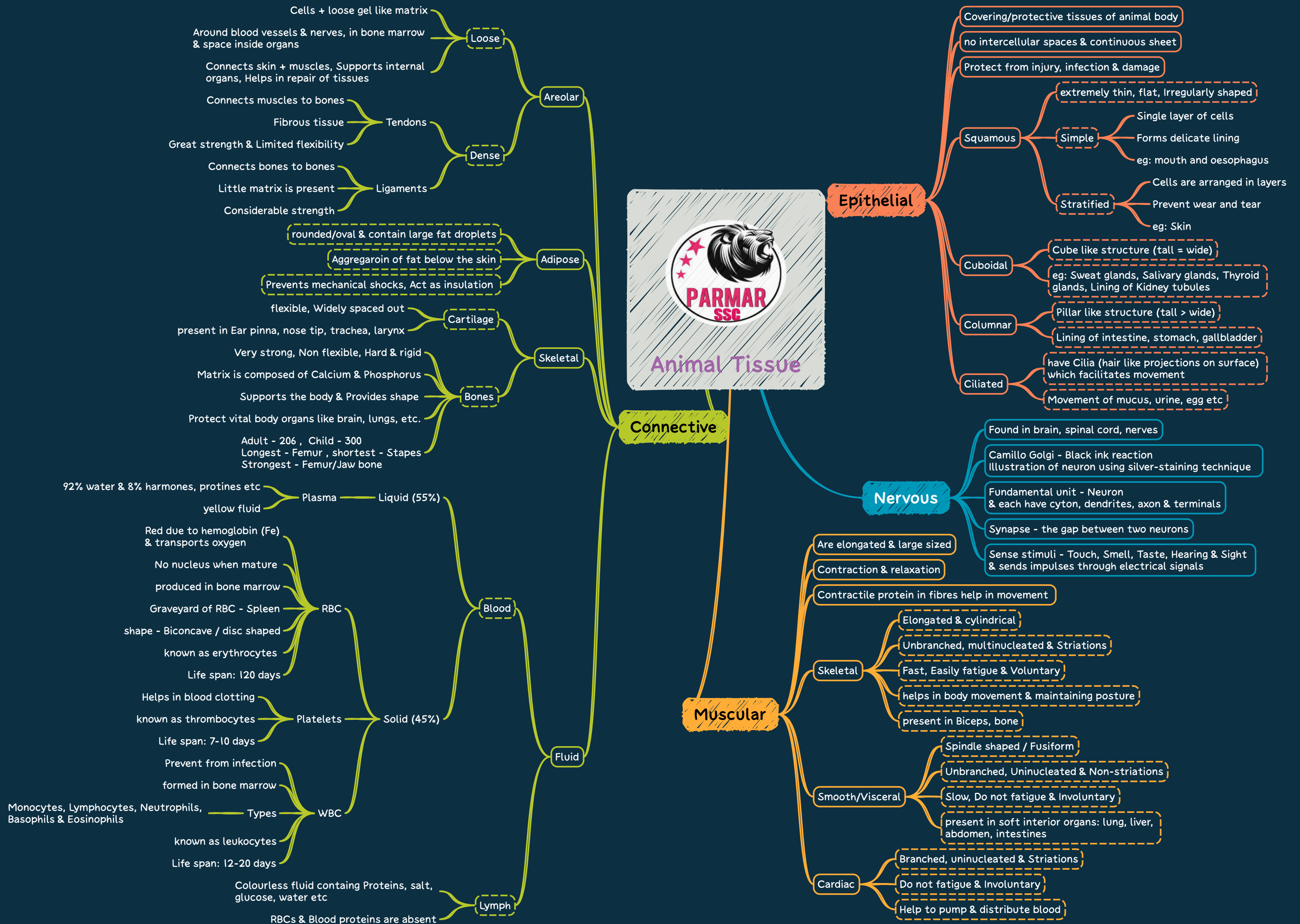
Cytoskeleton

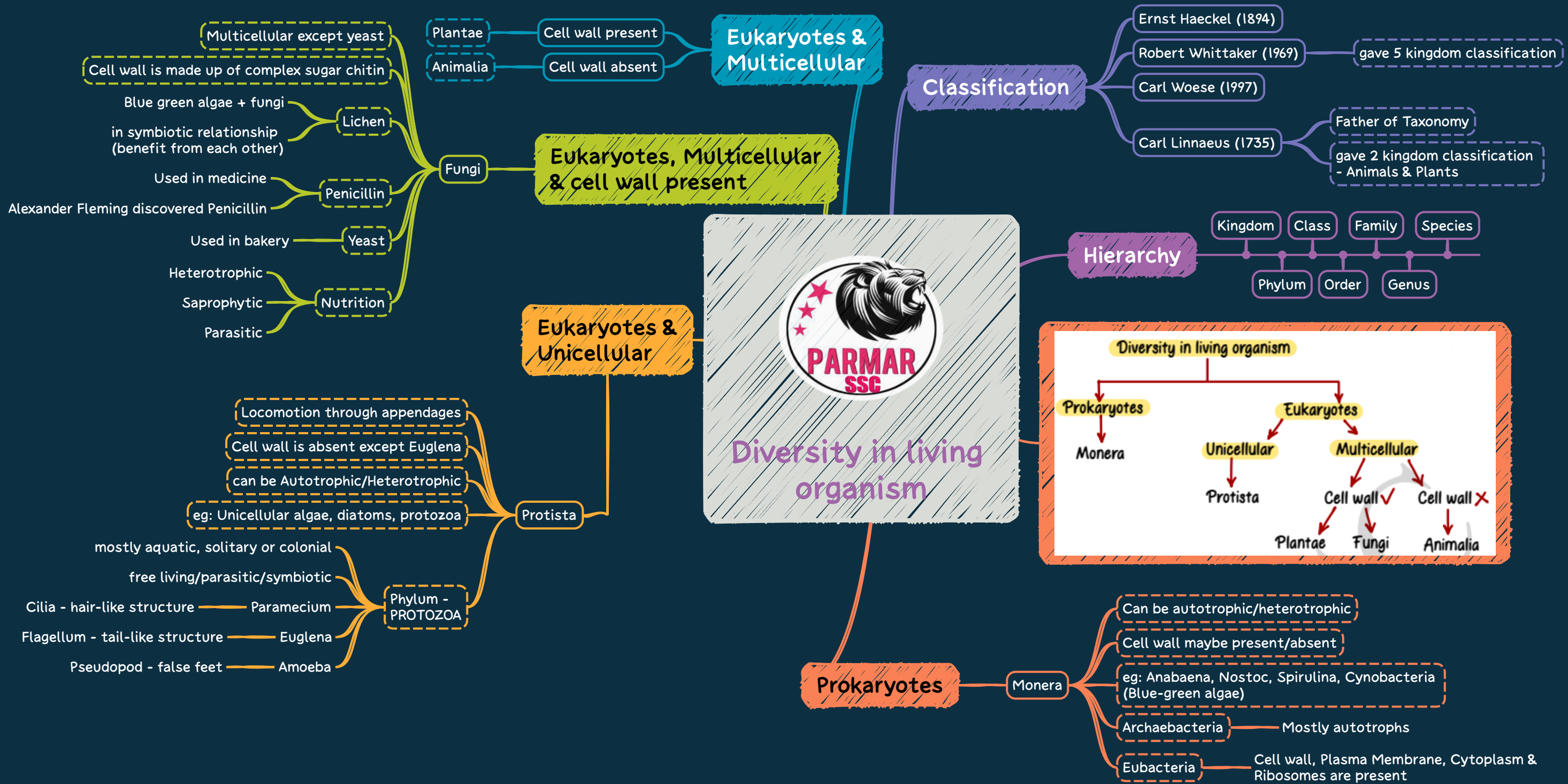
- Helps in providing structural support & movement to cell
- Made of two Centrioles in Cylindrical & 9 + 0 / 9 + 2 Form
- Microtubules organising centre

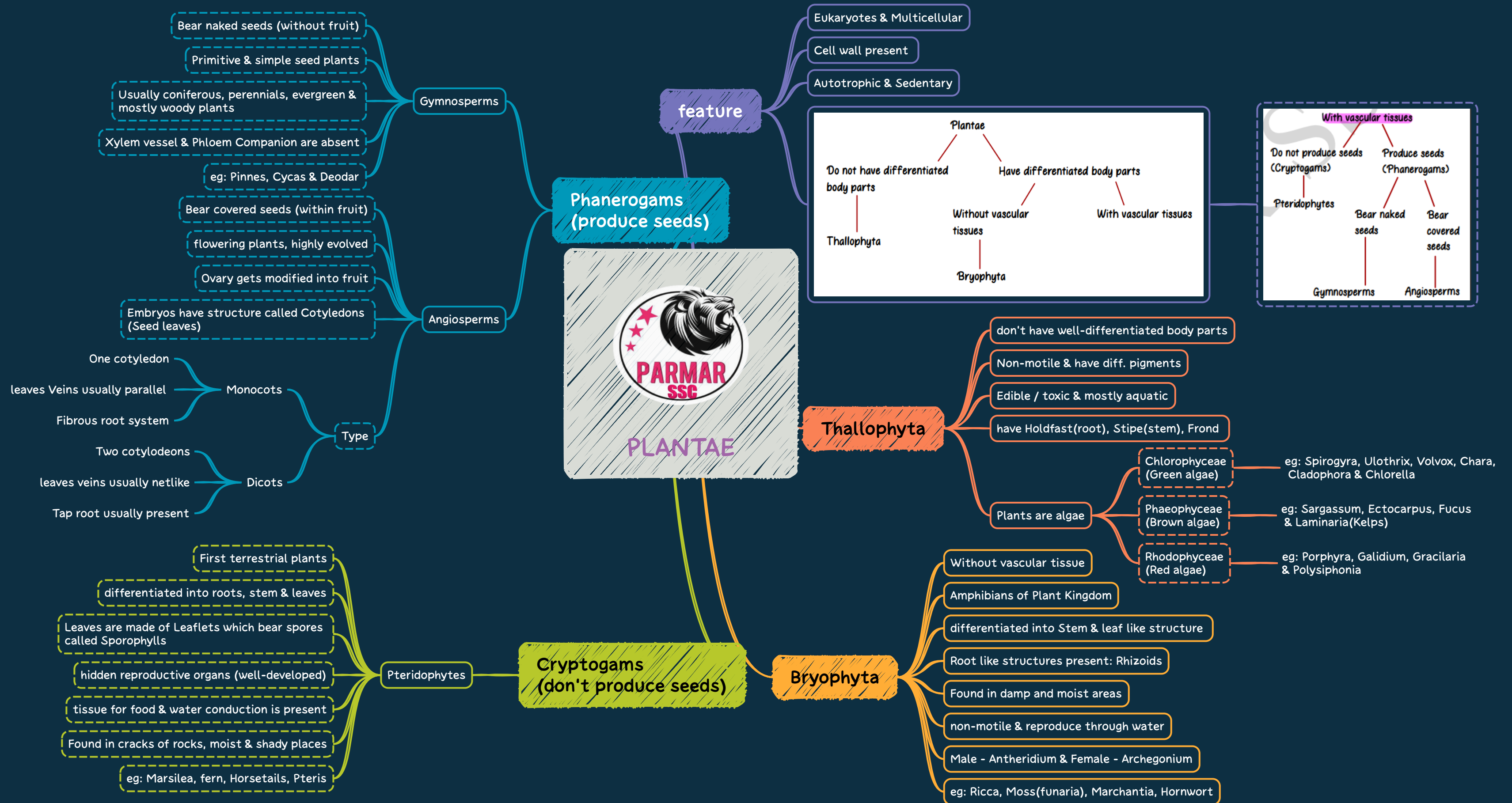
DNA Replication

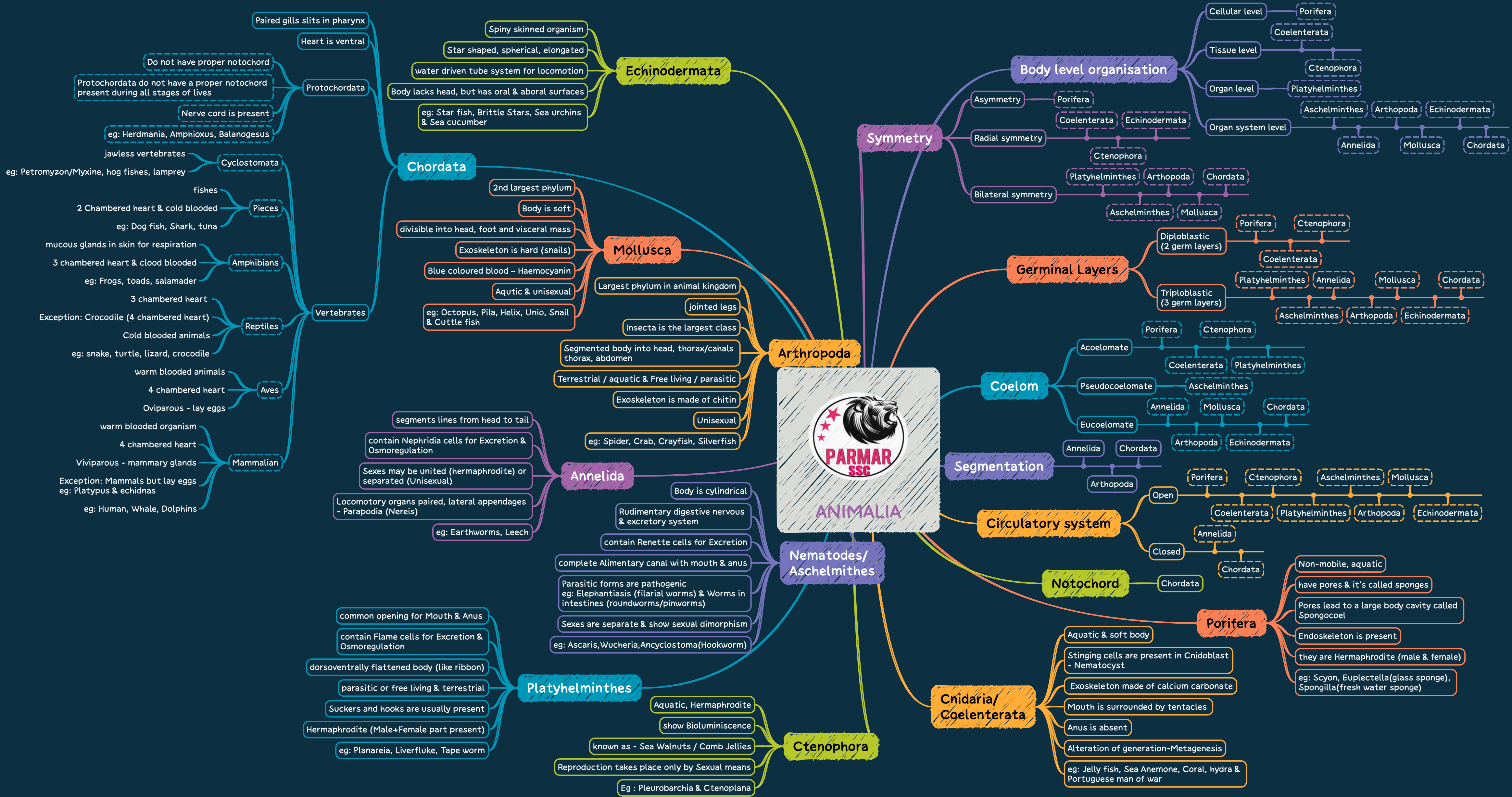
- cell copies its DNA before dividing, ensuring each new cell has identical copy of original DNA
- 4 base pairs (in DNA)
 - Adenine — Thymine
 - Guanine — Cytosine
- Base pairs (in RNA)
 - Adenine — Uracil
 - Guanine — Cytosine
- Enzymes essential For DNA replication in an animal cell
 - Topoisomerase: Prevents DNA from getting tangled (avoids supercoiling)
 - DNA Ligase: This enzyme facilitates the joining of DNA strands together
 - DNA Polymerase: This enzyme doubles the amount of DNA in a cell during cell division (builder)
 - DNA primase: It initiates DNA replication (new strands are produced)
 - Helicase: Unwinding enzyme

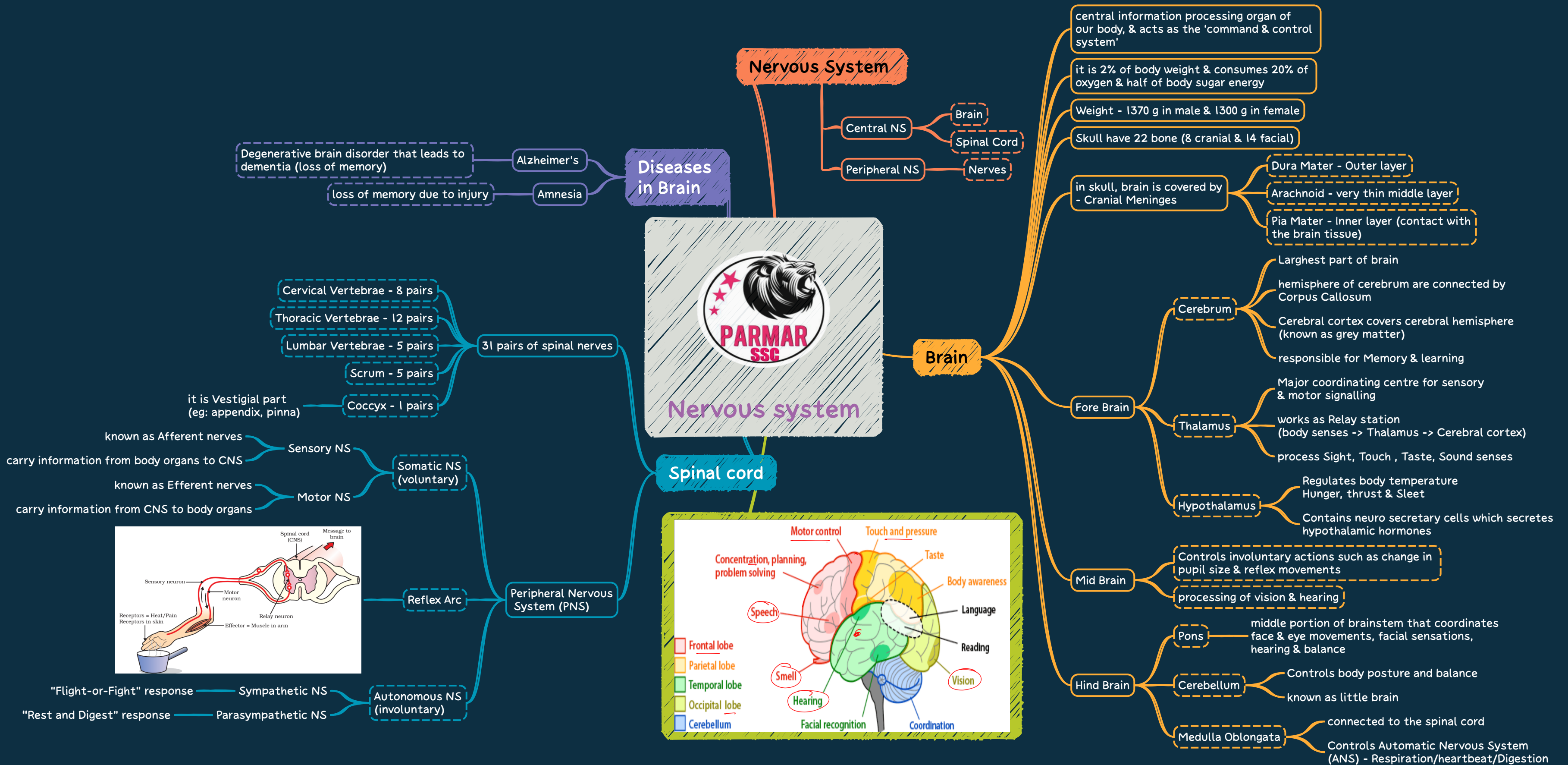
















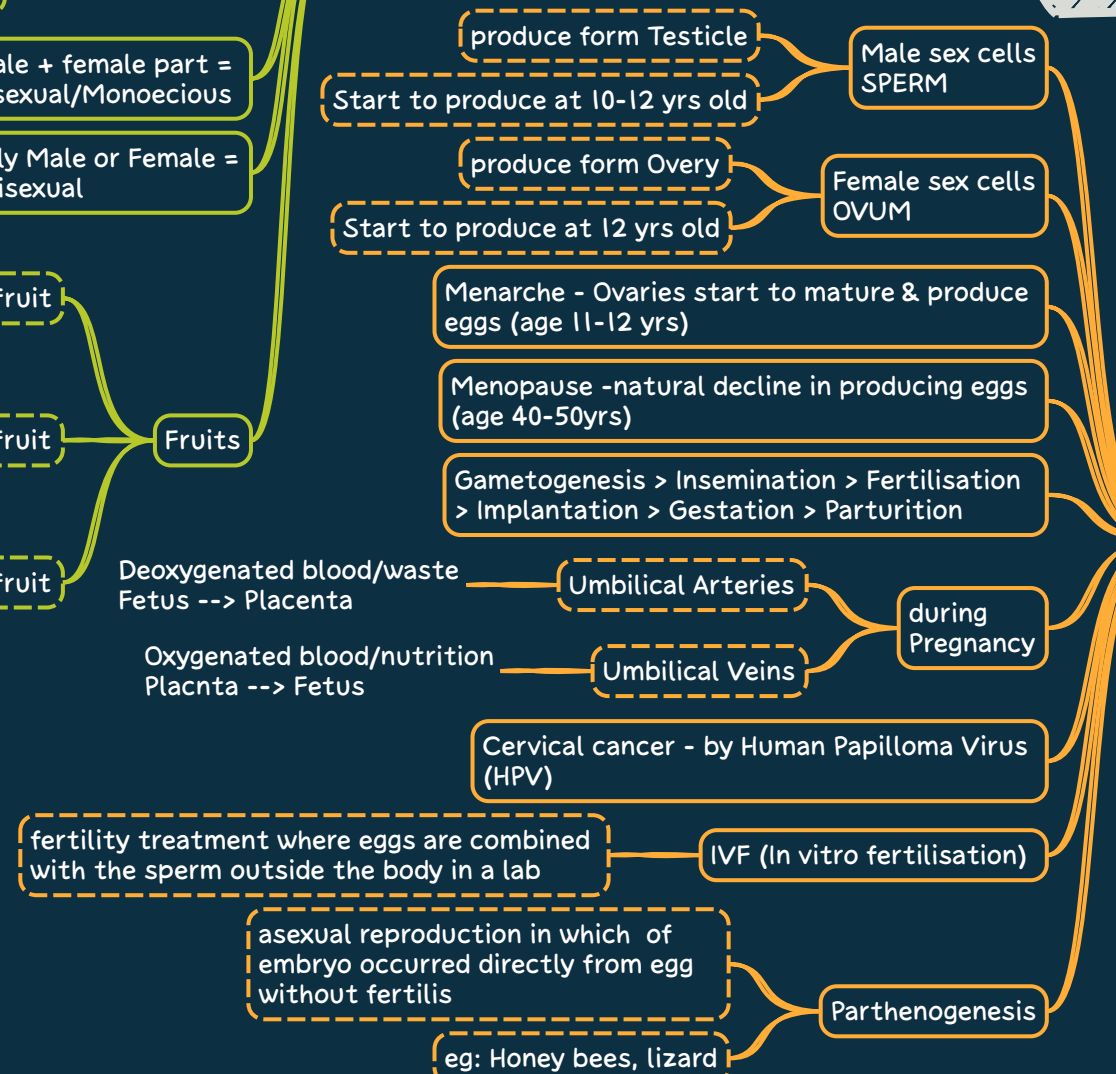
SEXUAL in PLANT

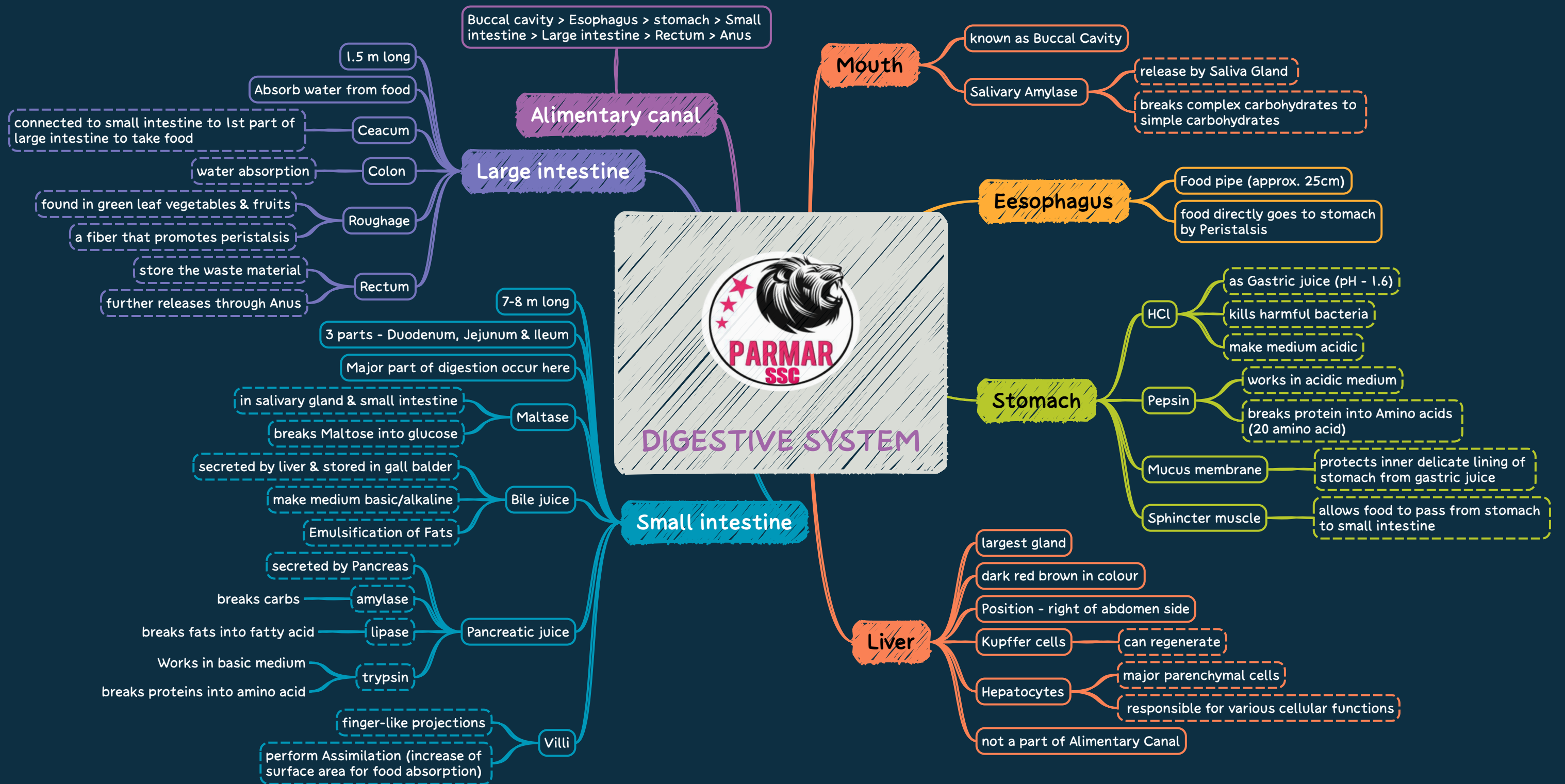


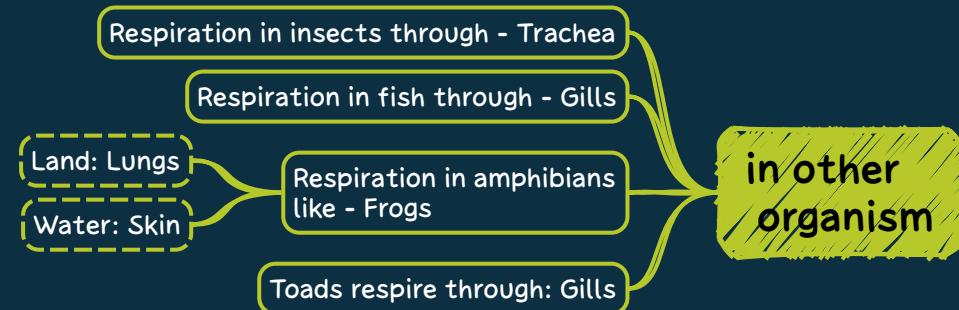
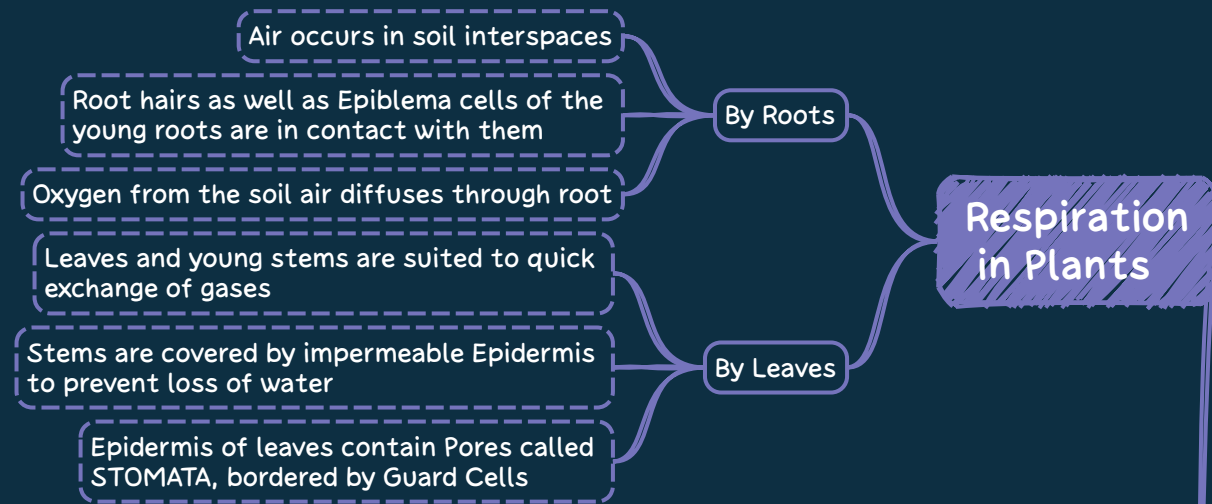
ASEXUAL



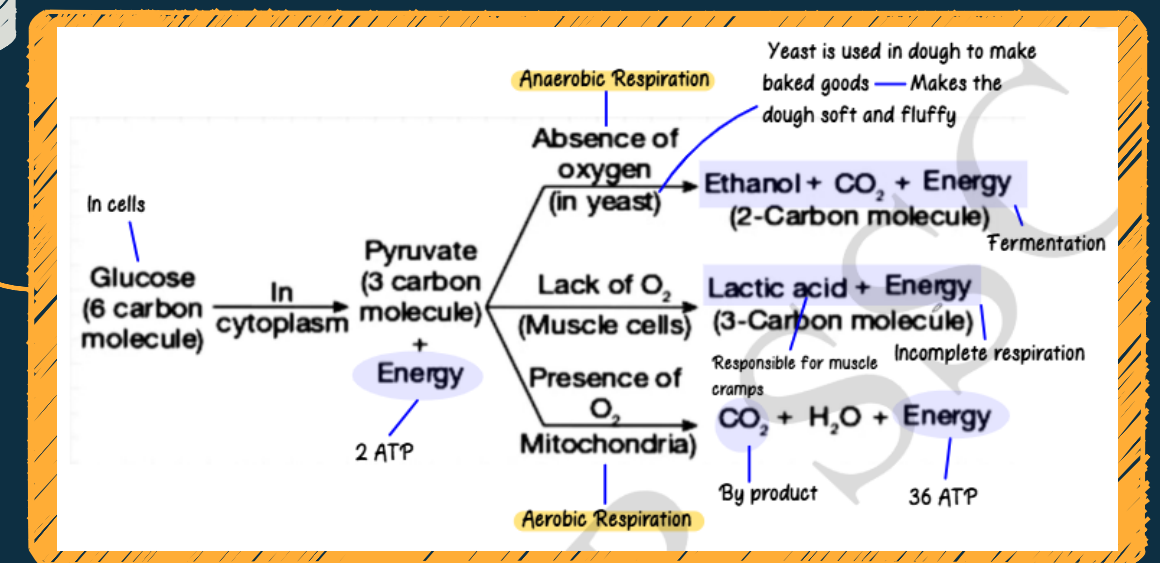
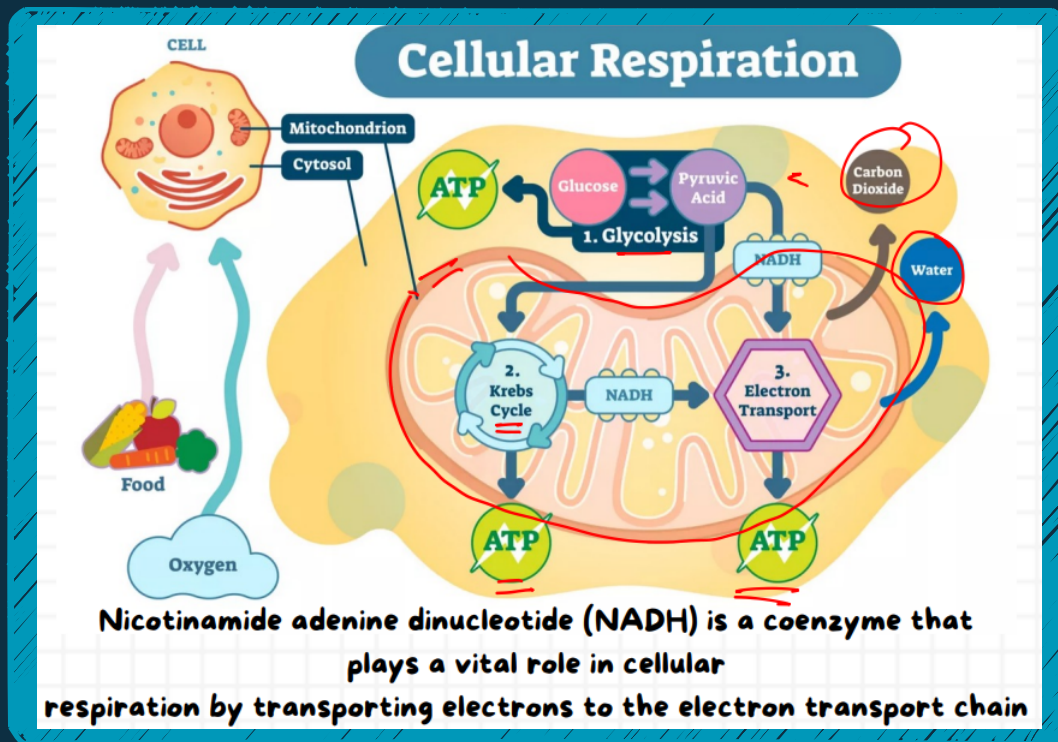
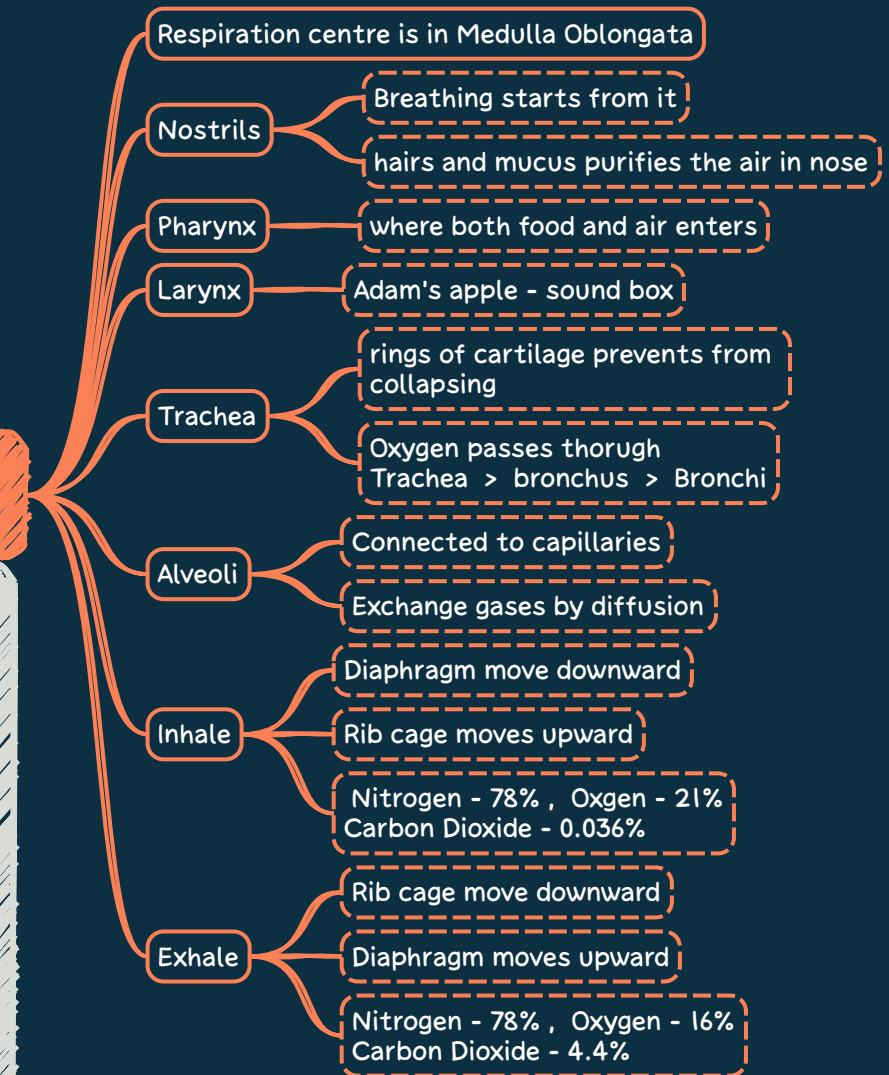
SEXUAL in HUMANS

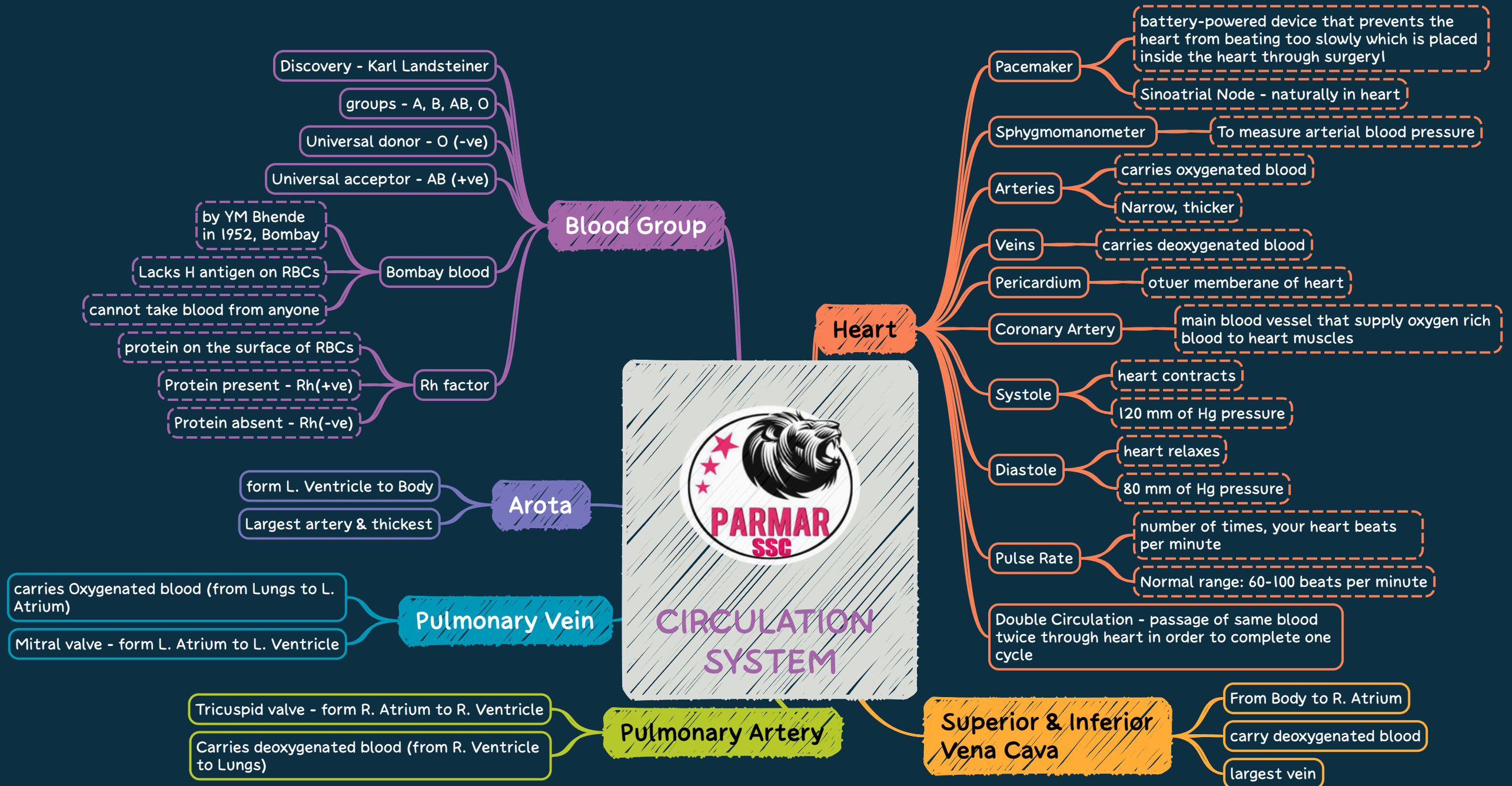


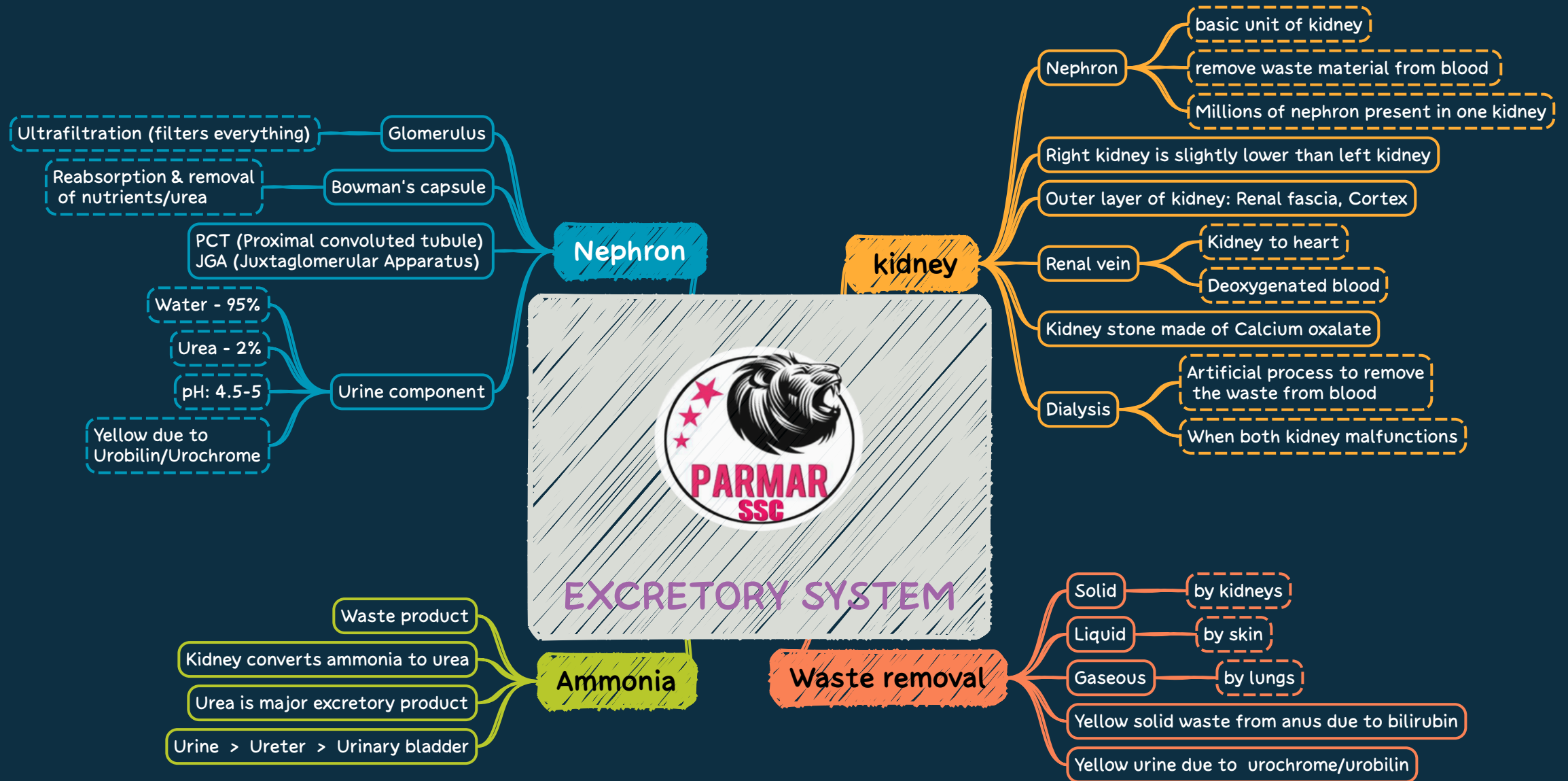


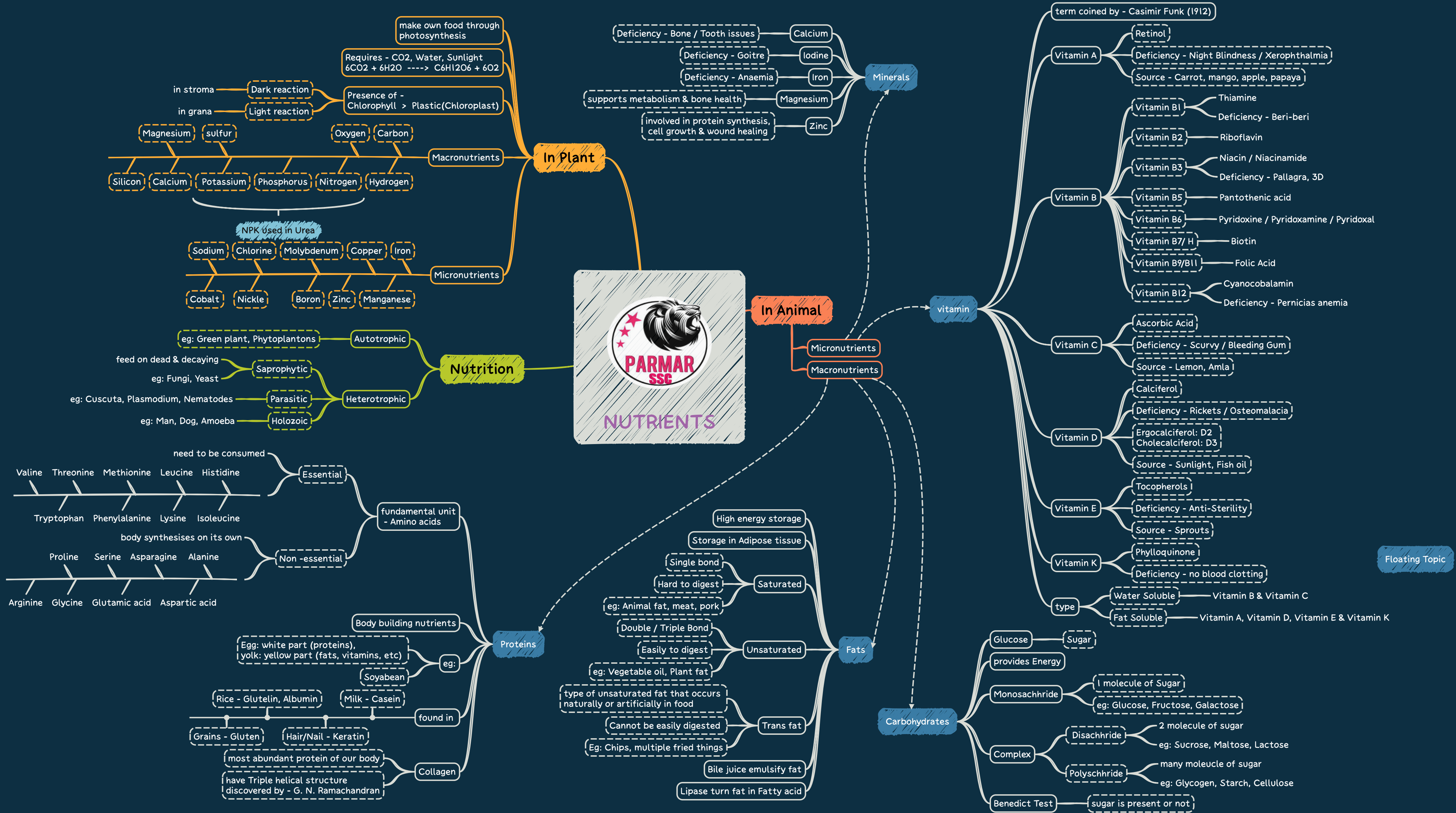


Respiration in animal









Floating Topic

